Manuscript Tables

Applying Causal Machine Learning and Multi-Omics Integration to Unravel Gene Regulatory Networks in Triple-Negative Breast Cancer (TNBC)

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Table 1. Overview of the single-cell CRISPRi perturbation dataset and cohort parameters in TNBC (GSE224986).

Metric / Parameter	Value / Description
Organism	Homo sapiens (Human)
Breast Cancer Subtype	Triple-Negative Breast Cancer (TNBC)
Cell Line	MDA-MB-231
CRISPR Perturbation Modality	CRISPR interference (CRISPRi / dCas9-KRAB)
Screen Type	Somatic Mutation Regulatory Enhancer Screen
Total Sequenced Cells	95,678
Total Filtered Genes	32,799
Total CRISPRi Guides Preserved	11,975
Target Enhancer Loci	819
Single-Guide Perturbation Cohort (Primary Causal Analysis)	3,930 cells
Multi-Guide Combinatorial Cohort	83,981 cells
Unassigned / Control Cohort	7,767 cells
Experimental 10x Genomics Lanes	8 lanes (Lanes 1-8)

Note: Single-cell libraries were generated using 10x Genomics Chromium 3' chemistry with direct guide capture across 8 lanes (Lewis et al., 2024). Full cell metadata also preserved in Supplementary Data 1.

Table 2. Top transcription factors physically bound to perturbed enhancers in TNBC (ENCODE ChIP-seq priors).

Transcription Factor	Target Enhancers	Total Binding Events	Mean ChIP Confidence	Max ChIP Confidence
CTCF	242	253	485.6	1000
POLR2A	238	249	511.4	1000
EP300	175	186	557.3	1000
JUND	155	166	517.9	1000
RAD21	148	158	406.3	1000
IKZF1	145	156	445.7	1000
SP1	142	149	545.1	1000
REST	142	149	323.9	1000
MYC	139	151	559.1	1000
FOS	137	142	553.4	1000
MAX	125	138	570.8	1000
YY1	123	128	470.5	1000

Note: Top 12 transcription factors ranked by target enhancer occupancy shown. Complete 15,059-edge prior matrix provided in Supplementary Table 1 (Supplementary_Table_1_Full_15k_TF_Enhancer_Priors.xlsx).

Table 3. Top master causal transcription factors prioritized in the triangulated TNBC regulatory network.

Transcription Factor	Controlled Enhancers	Target Genes	Total Causal Edges	Mean Causal Weight	Max Causal Weight
CTCF	106	2228	118738	2.30	42.42
POLR2A	101	2151	110463	2.54	50.84
JUND	69	2103	75316	2.49	45.33
EP300	65	2078	67275	2.85	32.42
IKZF1	58	2039	63658	2.02	29.38
RAD21	58	2041	63166	2.30	37.77
REST	57	2024	62513	1.53	32.76
MAX	52	2009	61079	2.56	40.82
MYC	54	2007	60903	2.92	29.59
SP1	56	2036	59043	2.72	38.42
SIN3A	50	1997	55113	3.11	34.04
FOS	52	2030	54762	2.84	47.84

Note: Top 12 causal regulators ranked by total causal outgoing edges shown. Complete multi-million causal edge matrix provided in Supplementary Data 3.

Table 4. Prioritized TNBC transcriptional master drivers with network centrality and phenotypic scores.

Driver TF	Target Genes	PageRank	Cumulative Weight	Upregulated	Downregulated	Master Score
CTCF	2228	0.000329	272,518.4	2201	27	0.9861
POLR2A	2151	0.000329	280,330.4	2124	27	0.9827
EP300	2078	0.000329	191,912.5	2051	27	0.8086
JUND	2103	0.000372	187,335.6	2076	27	0.8061
MYC	2007	0.000330	177,786.0	1980	27	0.7675
SIN3A	1997	0.000329	171,489.5	1970	27	0.7540
SP1	2036	0.000329	160,854.3	2009	27	0.7438
FOS	2030	0.000329	155,256.3	2003	27	0.7325
HDAC2	2003	0.000361	158,059.2	1976	27	0.7314
MAX	2009	0.000329	156,198.2	1982	27	0.7294

Note: Top 10 prioritized drivers ranked by Master Regulatory Score shown. Complete 25-driver cohort provided in repository table.

Table 5. Explainable AI (SHAP) feature attribution rankings and causal regulatory weights for master transcriptional drivers in TNBC.

Master Regulator	Mean Absolute SHAP Impact	Feature Weight Sum
JUND	0.029582	3,798.18
MYC	0.017427	3,391.99
MAX	0.012536	3,594.96
EP300	0.011668	3,690.95
CTCF	0.008296	4,484.27
HDAC2	0.006833	3,451.96
SIN3A	0.006764	3,713.95
SP1	0.005498	3,979.86
POLR2A	0.004045	4,186.99
FOS	0.003714	3,931.08

Note: SHAP values computed via TreeExplainer from XGBoost Regressors predicting target gene perturbation responsiveness (test R2 = 0.742).

Table 6. Quantitative benchmarking performance metrics across GRN reconstruction methods in TNBC single-cell perturbation data.

Method	AUROC	AUPRC	Precision (Top 25%)	Recall (Top 25%)	F1-Score	False Discovery Rate (FDR)
Triangulated Causal ML	0.8999	0.8745	0.9013	0.5633	0.6933	0.0987
ChIP-seq Prior Alone	0.8890	0.7228	0.7560	0.4725	0.5815	0.2440
Co-expression (Correlation)	0.7292	0.6814	0.6773	0.4233	0.5210	0.3227
GENIE3 (Tree Regression)	0.6698	0.6323	0.6427	0.4017	0.4944	0.3573

Note: Benchmarked against held-out experimental CRISPRi ground truth (|log2 FC| >= 0.50, |t| >= 3.0) and negative control edges in MDA-MB-231 cells.

Table 7. Actionable therapeutic targets, inhibitor classes, clinical trial phases, and predicted network sensitivity scores in TNBC.

Master Regulator	Target Mechanism	Candidate Inhibitors	Clinical Phase	Mechanism of Action	Sensitivity Score
MYC	Super-Enhancer / BRD4 Transcription	JQ1, OTX015 (Birabresib), THZ1 (CDK7i)	Phase I/II Clinical Trials	Disrupts BRD4 chromatin remodeling at oncogenic super-enhancers	0.94
EP300	Histone Acetyltransferase (HAT) Domain	A-485, CCS1477 (Inobrodib)	Phase I/II Clinical Trials	Blocks histone H3K27 acetylation, collapsing enhancer clusters	0.89
HDAC2	Class I Histone Deacetylase	Vorinostat (SAHA), Entinostat, Panobinostat	FDA Approved / Phase II in TNBC	Reverses epigenetic silencing of pro-apoptotic genes	0.85
JUND / FOS	AP-1 Dimerization & DNA Binding	T-5224, Verteporfin	Preclinical / Repurposed	Inhibits AP-1 DNA binding; suppresses EMT and metastasis	0.82
SP1	Zinc Finger Transcription Factor	Mithramycin A, EC-8042	Phase I / Preclinical	Binds GC-rich promoters, displacing oncogenic transcriptional initiation	0.78
MAX	MYC-MAX Dimerization Interface	10058-F4, MYCMI-6	Preclinical Evaluation	Directly disrupts heterodimerization with MYC to block target transactivation	0.74

Note: Predicted sensitivity scores derived from in silico topological network repression upon simulated target inactivation.